

freezing rain

Y18 (2018) — English translation

At the beginning of November 2016, precipitation in the form of freezing rain was observed in Moscow and the region, resulting from a temperature inversion, when the temperature at the Earth's surface is lower than at an altitude of 2-3 km. The characteristic dependences of temperature on height are shown in the figure. A “drop” of freezing rain near the ground is an icy spherical shell containing water inside

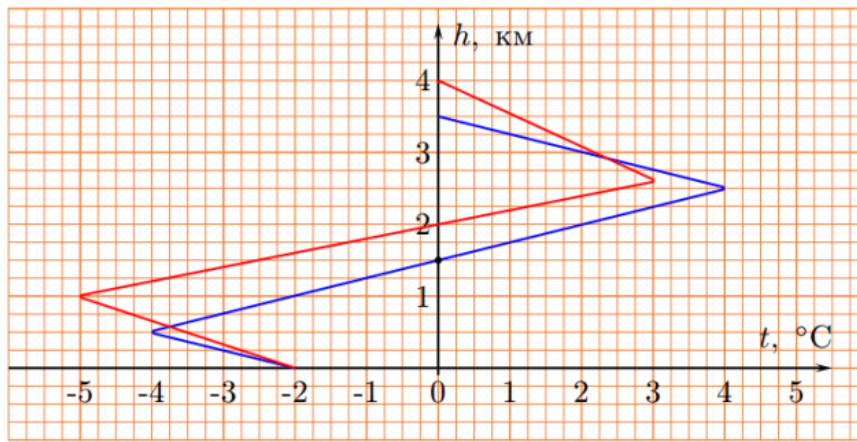


Figure 1: Figure 1.

A1 For a temperature profile reaching the temperature $+4^{\circ}\text{C}$, droplets of the maximum radius $R_0 = 1$ mm have time to freeze completely when flying towards the ground, but droplets of a larger size do not. What part of the mass of a drop of radius $R_1 = 2$ mm will freeze as it moves towards the ground? **8.00**

A2 Determine the limiting radius $R_0^{(1)}$ for the second temperature profile. **2.00**

Assume that at the altitude where the temperature passes through 0°C (for example, 1.5 km in the blue graph), all raindrops are liquid at a temperature $T_0 = 0^{\circ}\text{C}$. The heat removal from a unit surface of a droplet per unit time satisfies the empirical relation

$$\frac{\Delta Q}{\Delta S \Delta t} \sim \frac{v^{0.6}}{R^{0.3}} (T_d - T),$$

where T_d is the temperature of the droplet, T is the temperature of the surrounding air, v is the speed of the droplet, and R is the droplet radius. The drag force is proportional to the square of the speed and the square of the radius:

$$F_{\text{drag}} = v^2 R^2.$$

Evaporation and condensation, as well as changes in the radius of the drop due to crystallization, are neglected. Assume that the thermal conductivity of ice and water is high enough so that the temperature at the center of the droplet equals the temperature at its surface.

Source: <https://pho.rs/p/3044>